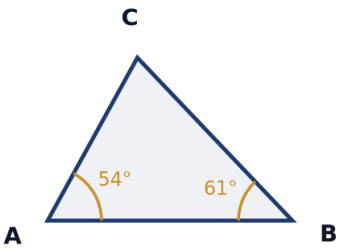
Proving triangles similar

Core Concepts: Proving triangles similar

1 In $\triangle ABC$ studying proving triangles similar, $m\angle A = 38^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 97°

b. 107°

c. 89°

d. 142°

2 A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm

c. 21 cm

d. 28 cm

3 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 8)$. What is its y-coordinate when $x = 1$?

a. 8

b. 13

c. 11

d. 14

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying proving triangles similar, $m\angle A = 38^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 90°

b. 100°

c. 82°

d. 142°

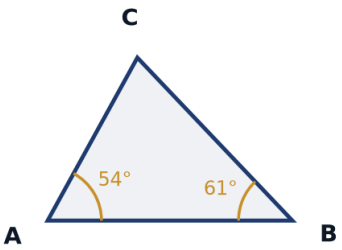
5 A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?

a. 18 cm

b. 21 cm

c. 28 cm

d. 63 cm



6 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$?

a. 11

b. 16

c. 14

d. 17

Graphical & Geometric Interpretation

7 In $\triangle ABC$ studying proving triangles similar, $m\angle A = 38^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 83°

b. 93°

c. 75°

d. 142°

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8 A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm

c. 35 cm

d. 112 cm

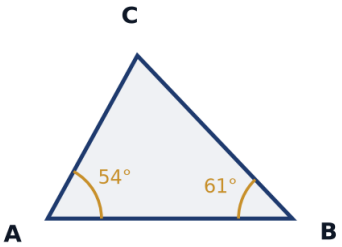
9 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 8)$. What is its y-coordinate when $x = 3$?

a. 14

b. 19

c. 17

d. 20



Exam-Style Problem Solving & Modeling

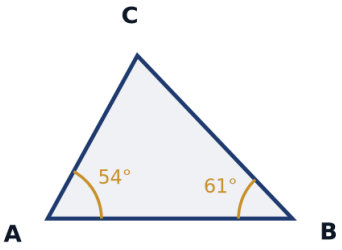
10 In $\triangle ABC$ studying proving triangles similar, $m\angle A = 38^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 76°

b. 86°

c. 68°

d. 142°



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11 A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm

c. 42 cm

d. 175 cm

12 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$?

a. 17

b. 22

c. 20

d. 23

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